Apples and Bananas

Time limit: 1.00 s **Memory limit:** 512 MB

There are n apples and m bananas, and each of them has an integer weight between $1 \ldots k$. Your task is to calculate, for each weight w between $2 \ldots 2k$, the number of ways we can choose an apple and a banana whose combined weight is w.

Input

The first input line contains three integers k, n and m: the number k, the number of apples and the number of bananas.

The next line contains n integers $a_1, a_2, \ldots, a_n$: weight of each apple.

The last line contains m integers $b_1, b_2, \ldots, b_m$: weight of each banana.

Output

For each integer w between $2 \ldots 2k$ print the number of ways to choose an apple and a banana whose combined weight is w.

Constraints

- $1 \le k, n, m \le 2 \cdot 10^5$
- $1 \le a_i \le k$
- $1 \le b_i \le k$

Example

Input	Output

`5 3 4`
`5 2 5`
`4 3 2 3`
	`0 0 1 2 1 2 4 2 0`

Explanation: For example for $w = 8$ there are 4 different ways: we can pick an apple of weight 5 in two different ways and a banana of weight 3 in two different ways.